Developmental Motor Coordination in RF-Shielded Versus Standard Electromagnetic Environments

A Controlled Mouse Study Proposal

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1. Purpose

HOWL-NEURO-1-2026 establishes by carrier elimination that electromagnetic field propagation is the only known physical mechanism with sufficient speed to coordinate sub-millisecond motor timing across biological distances. The carrier elimination is arithmetic: distance divided by speed equals transit time, transit time compared to the control theory delay ceiling equals pass or fail. Every classical carrier fails for demanding motor tasks. Only EMF passes, with six orders of magnitude of headroom.

This paper proposes the experiment that tests the developmental consequence. If endogenous EMF coordination exists and operates at femto-Tesla amplitudes, then chronic exposure to exogenous RF at milliwatt power densities during the developmental window — when motor coordination networks are being established for the first time — may degrade the precision of those networks.

The experiment raises two populations of mice from conception in identical conditions except electromagnetic environment. One population develops in an RF-shielded facility. The other develops in a standard facility with continuous WiFi-equivalent RF exposure. Both populations are tested on a battery of motor, sensory, and cognitive-motor tasks spanning the demand spectrum from undemanding to maximally demanding.

The primary hypothesis: shielded mice will show measurably faster reaction times, more precise inter-muscle timing, higher coordination ceilings, and faster sensory-motor learning than RF-exposed mice, with the difference increasing as task demand increases. Undemanding tasks will show no difference, confirming that the effect is coordination-specific, not a general health or fitness effect.

This paper is a proposal. The author will not run the experiment. The proposal is published so that any researcher with access to a behavioral neuroscience facility and an

RF-shielded room can execute it. The experimental design, analysis plan, predictions, and falsification criteria are specified completely. Nothing is left to the implementer's judgment except the practical logistics of animal husbandry and equipment setup.

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2. The Developmental Timing Argument

The expanded Hebb's rule from HOWL-NEURO-1-2026 states: neurons that fire together first phase-lock through EMF coupling, then wire together through synaptic strengthening. The field is the scout. The synapse is the road crew. The scout establishes the coordination pattern. The road crew builds the permanent pathway.

In an RF-quiet environment, the scout operates in silence. Phase-lock between coordination nodes forms at the precision the endogenous field supports. The permanent synaptic pathways are built on a foundation of precise timing relationships.

In an RF-saturated environment, the scout operates in noise. The endogenous coordination signal at femto-Tesla levels competes with exogenous RF at milliwatt levels. The signal-to-noise ratio for establishing initial phase-lock is degraded. The coordination patterns that form are less precise. The permanent synaptic pathways are built on a noisier foundation.

The critical observation: mature coordination networks in adult animals may tolerate RF noise because the phase-lock relationships and synaptic pathways are already established. The dissolution to Zero-absent (HOWL-MATH-14-2026) has already occurred. The validity envelope of the dissolved coordination skill may be wide enough to accommodate the noise. But developing networks have no prior dissolution. Every coordination pattern is being established for the first time. The first phase-lock is the most vulnerable to noise because it has no prior structure to stabilize it.

This predicts that the developmental window is the critical exposure period. Animals raised in RF noise and then moved to a quiet environment may partially recover — the foundation is degraded but the system can still build on it. Animals raised in quiet and then exposed to RF as adults may show no degradation — the foundation is solid and the noise falls within the dissolution validity envelope.

The experiment tests the first prediction: developmental exposure produces measurable coordination differences that persist into adulthood.

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3. Study Design

3.1 Animals

120 C57BL/6J mice, 60 per group, balanced sex ratio. C57BL/6J is selected for extensive baseline motor characterization in the published literature, providing normative reference data for every task in the battery.

Bred in-house. Dams are assigned to groups before mating so that prenatal development occurs entirely in the assigned electromagnetic environment. The embryonic period is included in the developmental exposure because neural development begins before birth. Assigning dams after birth would miss the prenatal window.

3.2 Facility

Population A (shielded). Purpose-built RF-shielded room. Walls, floor, and ceiling lined with copper mesh providing minimum 80 dB attenuation at 2.4 GHz and 5 GHz. Ventilation through waveguide-below-cutoff ducts that pass air but block RF. Lighting, temperature control, and monitoring equipment are hardwired with filtered power lines. No wireless devices inside the enclosure. Ambient RF verified below 1 pW/m^2 by spectrum analyzer at daily intervals. All monitoring via shielded wired cameras.

Population B (exposed). Identical room dimensions, identical cage types, identical enrichment, identical lighting, temperature, humidity, and noise environment. Standard laboratory electromagnetic environment plus one commercial WiFi access point (2.4 GHz, 802.11n, 100 mW transmit power) positioned 2 meters from the cage rack, broadcasting continuously with standard beacon frames. Power density at cage level verified at 10-50 μW/m^2, representative of typical residential WiFi exposure. This is well within regulatory safety limits in all jurisdictions.

3.3 Control of non-electromagnetic variables

Identical diet: standard chow, same lot number, same delivery schedule. Identical water: same source, same filtration. Identical cage dimensions, bedding material, and change schedule. Identical enrichment: running wheel, nesting material, tunnel — same manufacturer, same dimensions. Identical handling: same handlers, same schedule, same gloves, same duration. Identical light-dark cycle: 12:12, same onset time. Identical temperature: 22 $\pm$ 1°C. Identical humidity: 45 $\pm$ 5%. Identical acoustic environment: verified by sound level meter. The RF shielding does not significantly attenuate acoustic frequencies.

3.4 Blinding

Testing personnel are blinded to group assignment. Mice are transported from their home room to a shared testing room in identical opaque carriers. The testing room has standard laboratory electromagnetic environment for both groups. The shielded group is exposed to standard RF only during the testing period, approximately 30-60 minutes per session. This controls for acute RF effects during testing. The variable under test is chronic developmental exposure, not acute exposure during performance.

3.5 Testing timepoints

Six developmental timepoints: postnatal day 14 (pre-weaning, eyes just opened), 21 (weaning), 28 (juvenile), 42 (adolescent), 56 (young adult), 84 (adult). The full battery

is administered at each timepoint to track developmental trajectory, not just endpoint performance.

3.6 Sample size justification

60 mice per group provides 80% power to detect a 0.5 standard deviation difference in inter-muscle timing precision at alpha = 0.05 (two-tailed), assuming 20% attrition over the 84-day study period. This is a conservative effect size. If the effect exists and is smaller than 0.5 SD, this study will not detect it, and that null result is reported as a finding with the confidence interval bounding the maximum compatible effect size.

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4. Testing Battery

The battery is organized in four tiers by coordination demand. Tier 1 tasks are controls that should show no group difference. Tier 2 tasks approach the classical carrier ceiling. Tier 3 tasks exceed the classical carrier ceiling. Tier 4 tasks test sensory-motor integration, learning speed, and attention switching, which depend on the timing channel for different reasons than pure motor coordination.

4.1 Tier 1 — Undemanding motor tasks (control tier)

These tasks are below the classical carrier ceiling at measured speeds. Both groups should perform identically. Any group difference on Tier 1 tasks indicates a general health, fitness, or motivation effect rather than a coordination-specific effect, and the timing channel hypothesis is not supported as the explanation.

Task 1.1: Open field locomotion. Mouse placed in standard open field arena (40 $\times$ 40 cm) for 10 minutes. Measures: total distance traveled, average speed, time in center zone (anxiety proxy), number of rearing events. Coordination demand: minimal, basic quadrupedal walking on flat surface.

Task 1.2: Wide beam traversal (2 cm width). Three traversals of a 1-meter beam. Measures: time to cross, number of foot faults, number of pauses. Coordination demand: low, four contact points on a wide surface with low center of mass.

Task 1.3: Simple reaction time. Light stimulus followed by food reward at a fixed location. Mouse must move from a start position to the reward location after the light. Measures: latency from light onset to movement onset (reaction time), latency from light onset to reward retrieval (total response time). 20 trials per session after 3 days of habituation training. Coordination demand: minimal, the task is a simple detection-response chain.

4.2 Tier 2 — Moderate motor tasks (boundary tier)

These tasks approach the classical carrier limit. Group differences may or may not appear.

Task 2.1: Narrow beam traversal (0.5 cm width). Three traversals of a 1-meter beam. Measures: time to cross, foot faults, falls, pauses. Coordination demand: moderate, reduced base of support increases balance correction frequency.

Task 2.2: Rotorod at fixed speed (15 RPM). Three trials of 300 seconds maximum. Measures: latency to fall. Coordination demand: moderate, continuous adjustment to curved moving surface at a constant rate.

Task 2.3: Choice reaction time. Light stimulus appears at one of two locations (left or right). Food reward is delivered at the illuminated location. Mouse must discriminate location and move to the correct side. Measures: reaction time, choice accuracy, total response time. 30 trials per session, 15 per side, pseudorandomized. Coordination demand: moderate, the task adds a discrimination step before the motor response.

4.3 Tier 3 — Demanding motor tasks (above classical ceiling)

These tasks exceed the classical carrier ceiling even at 4x extreme generosity. The timing channel hypothesis predicts group differences here.

Task 3.1: Accelerating rotorod (4-80 RPM over 300 seconds). Three trials. Measures: latency to fall, RPM at fall. Coordination demand: high, the progressive acceleration continuously increases the correction frequency required. The RPM at fall is a direct measure of the animal's coordination ceiling.

Task 3.2: Bipedal reach. Food pellet mounted above cage height. Mouse must stand on hind paws and reach upward. Measures: hold time per reach (time in sustained bipedal stance), maximum hold time across 10 reaches per session, number of successful pellet retrievals. Coordination demand: maximum in the standard mouse repertoire. Digitigrade posture, zero passive stability, full muscular suspension of the entire kinetic chain.

Task 3.3: Landing from 30 cm drop onto force plate. Three drops per session. Mouse is placed on a platform 30 cm above a force plate. The platform is withdrawn. Measures: peak ground reaction force, time from impact to stabilization (defined as force variation below 5% of body weight), bilateral limb loading symmetry (ratio of force under left versus right hind limbs). Coordination demand: high, impact absorption requires sub-millisecond inter-limb coordination.

Task 3.4: Rapid reversal. Mouse runs down a 1-meter beam toward a food reward. At a random point during the approach, a puff of air is delivered from the reward end, requiring the mouse to stop and reverse direction. Measures: reversal latency (time from air puff to direction change), distance traveled past the puff point before reversal (overshoot distance). Coordination demand: high, the task requires detecting the perturbation, inhibiting the current motor program, and initiating a new one within a tight temporal window.

4.4 Tier 4 — Sensory-motor integration and learning

These tasks test the timing channel through a different mechanism. Rather than pure motor coordination, they test how quickly the animal detects a change in its environment,

how quickly it adapts to a new contingency, and how precisely it coordinates attention switches between competing demands.

Task 4.1: Sensory detection threshold. A vibrotactile stimulus is delivered to the platform the mouse stands on. Stimulus amplitude decreases across trials using a staircase procedure. Measures: detection threshold (minimum amplitude producing a measurable orienting response), reaction time at each amplitude. The timing channel hypothesis predicts that the shielded group will have lower detection thresholds and faster reaction times because the sensory transduction chain includes timing-dependent components that are more precisely calibrated in the absence of developmental RF noise.

Task 4.2: Contingency learning speed. Phase 1: mouse learns that pressing lever A produces food. Measure: trials to criterion (8 of 10 correct). Phase 2: contingency reverses, lever B now produces food, lever A produces nothing. Measure: trials to criterion on reversal. Phase 3: second reversal back to lever A. Measure: trials to criterion. The timing channel hypothesis predicts faster learning and faster reversal in the shielded group because the sensory-motor loops that detect and respond to contingency changes operate through timing-dependent pathways.

Task 4.3: Serialized attention switching (wheel-food-wheel-food). The mouse has access to a running wheel and a food dispenser. The protocol introduces a pattern: after 2 minutes of wheel access, the wheel locks and a food pellet is dispensed. After 60 seconds of food access, the food port closes and the wheel unlocks. The cycle repeats. Measures: latency from wheel lock to food port approach on each cycle (how quickly the mouse detects the transition), latency from food port close to wheel re-engagement on each cycle, number of cycles to stable anticipatory behavior (the mouse moves to the food port before the pellet is dispensed, indicating it has learned the pattern), timing precision of anticipatory responses (how close to the transition point the anticipatory movement begins).

This task tests the timing channel at the learning level. The mouse must detect a temporal pattern, encode it, and produce anticipatory behavior timed to the pattern. The precision of the anticipatory timing is a direct measure of temporal encoding precision. If the timing channel is degraded by developmental RF exposure, temporal pattern learning will be slower and anticipatory timing will be less precise.

Prediction for Task 4.3: The shielded group will show faster detection of the transition (shorter latency from event to response), faster learning of the pattern (fewer cycles to anticipatory behavior), and more precise anticipatory timing (anticipatory movements begin closer to the transition point with less trial-to-trial variability). The exposed group will eventually learn the same pattern but will take more cycles to reach criterion, and the anticipatory timing will have higher variability even after learning is complete.

Task 4.4: Complex reaction time with sequential demands. A three-stimulus sequence is presented: tone (orient), light at one of three locations (discriminate), and the mouse must move to the lit location and press a lever within a response window. The sequence repeats with variable inter-trial intervals. Measures: reaction time to each component (tone detection, light discrimination, lever press), total sequence completion time, accuracy, and anticipatory errors (pressing before the light).

The task demands attention switching across three modalities (auditory, visual, motor) in a fixed sequence. The timing channel hypothesis predicts that the shielded group will show tighter temporal coupling between the three components — the transition from hearing the tone to orienting to the light to executing the motor response will be faster and more consistent, because each transition involves a timing-dependent handoff between neural subsystems.

Task 4.5: Contextual fear conditioning — timing precision. Standard contextual fear conditioning with a twist: the conditioned stimulus (a tone) is paired with a mild foot shock at a fixed delay (e.g., 30 seconds after tone onset). After conditioning, the mouse is placed in the context and the tone is presented without shock. Measures: onset latency of freezing response (time from tone onset to first freeze), and critically, the temporal distribution of freezing relative to the expected shock time. A mouse that has precisely encoded the tone-shock interval will show peak freezing at 30 seconds after tone onset. A mouse with less precise temporal encoding will show broader freezing that begins earlier and extends later.

The timing channel hypothesis predicts sharper temporal discrimination in the shielded group — freezing response more tightly clustered around the learned interval, with less temporal spread.

4.5 EMG recording

Surface EMG electrodes are placed bilaterally on the gastrocnemius (ankle extensors) during all Tier 2, Tier 3, and selected Tier 4 tasks. Sampling rate: 10 kHz. The primary EMG measure is bilateral onset latency difference — the time between EMG onset in the left and right gastrocnemius during bilateral activation events (landing impact, rotorod step recovery, bipedal reach initiation, rapid reversal). The standard deviation of bilateral onset latency across trials within a session is the measure of inter-muscle timing precision.

EMG is not recorded during Tier 1 tasks. If Tier 1 tasks show group differences, the study has a confound and EMG analysis of higher tiers is uninterpretable.

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5. Predictions

5.1 The demand gradient

The central prediction is a demand gradient — group differences that increase with task demand.

Tier	Tasks	Predicted group difference
1	Open field, wide beam, simple RT	None. Both groups identical.
2	Narrow beam, fixed rotorod, choice RT	Small or absent. May not reach significance.

Tier	Tasks	Predicted group difference
3	Accel. rotorod, bipedal reach, landing, rapid reversal	Measurable. Shielded group shows better timing precision, higher coordination ceiling.
4	Sensory detection, learning speed, attention switching, complex RT, temporal conditioning	Measurable. Shielded group shows faster detection, faster learning, more precise temporal encoding.

If the demand gradient is absent — if group differences are uniform across tiers, or present only at Tier 1 — the timing channel hypothesis is not supported.

5.2 Specific task predictions

Simple reaction time (Task 1.3). No group difference. This is the control for reaction time tasks. If the shielded group has faster simple reaction times, the difference in more complex reaction times could be explained by a general speed advantage rather than a timing-specific advantage. For the hypothesis to be supported, simple reaction time must be equivalent, and the group difference must emerge only when the task adds coordination, discrimination, or temporal encoding demands.

Choice reaction time (Task 2.3) versus simple reaction time (Task 1.3). The difference between choice RT and simple RT isolates the discrimination component. If the shielded group shows a larger advantage on choice RT than on simple RT, the discrimination process itself benefits from better timing. If both show the same group difference (or no difference), the timing channel is not involved in discrimination.

Accelerating rotorod (Task 3.1). RPM at fall is the coordination ceiling. The shielded group is predicted to fall at a higher RPM, indicating a higher maximum coordination frequency. The difference in RPM at fall, multiplied by $2\ \pi$, gives an estimate of the coordination frequency advantage in Hz, which can be compared to the theoretical predictions from HOWL-NEURO-1-2026.

Bipedal reach hold time (Task 3.2). The shielded group is predicted to maintain bipedal stance longer. The hold time distribution (across 10 reaches per session) is predicted to have a higher mean and lower variance in the shielded group, indicating both better sustained coordination and more consistent coordination.

Landing bilateral symmetry (Task 3.3). The ratio of left to right hind limb force during landing is predicted to be closer to 1.0 in the shielded group, indicating more precise bilateral timing of impact absorption. The exposed group is predicted to show greater asymmetry and greater trial-to-trial variability in the symmetry ratio.

Wheel-food-wheel-food anticipatory timing (Task 4.3). This is the task most directly sensitive to the developmental timing hypothesis. The temporal pattern (2 minutes wheel,

60 seconds food, repeat) must be encoded by the animal's timing circuitry. The shielded group is predicted to reach anticipatory behavior in fewer cycles and to show tighter temporal precision in the anticipatory response. The exposed group is predicted to learn the pattern eventually but with more cycles required and broader temporal distribution of anticipatory responses.

The standard deviation of anticipatory timing across cycles after learning criterion is reached is the purest measure of temporal encoding precision in the battery. If this measure shows a group difference, it directly supports the hypothesis that developmental RF exposure degrades temporal precision.

Contextual fear conditioning temporal precision (Task 4.5). The width of the freezing distribution around the expected shock time is the measure. The shielded group is predicted to show a narrower distribution — more precise temporal memory of the conditioned interval. The exposed group is predicted to show a broader distribution — less precise temporal encoding of the same interval despite equivalent conditioning.

5.3 Developmental trajectory predictions

Both groups improve with age on all tasks. The predictions concern the rate and ceiling of improvement.

Rate. The shielded group is predicted to reach performance plateaus earlier — steeper improvement curves from P14 to P42. The exposed group catches up partially but reaches plateau later.

Ceiling. The shielded group is predicted to reach a higher performance plateau on Tier 3 and Tier 4 tasks. The ceiling difference is predicted to be stable from P56 to P84, indicating that it is not a developmental delay (which would close with time) but a permanent calibration difference.

Tier interaction. The ceiling difference is predicted to be absent or negligible on Tier 1 tasks, small on Tier 2 tasks, and largest on Tier 3 and Tier 4 tasks. This demand-dependent ceiling difference is the signature of a coordination-specific effect rather than a general developmental effect.

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6. Falsification Criteria

F1: Tier 1 performance differs between groups. Any significant group difference on open field locomotion, wide beam traversal, or simple reaction time indicates a general health, fitness, or motivation effect. The timing channel hypothesis is not the explanation. The study has a confound.

F2: Tier 3 and Tier 4 timing precision is identical between groups at all developmental timepoints. No group difference on any demanding task at any age. The chronic RF exposure has no measurable effect on motor timing development. The timing channel

either does not exist, is robust to 2.4 GHz RF at residential power levels, or the effect size is below the study's detection threshold. This is the cleanest falsification.

F3: Group difference is uniform across tiers. The shielded group outperforms on all tasks including Tier 1 by approximately equal margins. The effect is general, not coordination-specific. The timing channel hypothesis is not supported as the mechanism. The difference may be attributable to uncontrolled variables in the shielded versus exposed environment.

F4: Timing precision differs but task performance does not. Better bilateral onset latency precision in the shielded group does not translate to better hold times, later falls, or faster learning. Timing precision is not the rate-limiting factor for these tasks. The timing channel may exist but is not the constraint predicted by the demand matrix.

F5: The developmental trajectory is identical between groups. Same slope, same plateau, same shape. Chronic RF exposure during development has no effect on the rate or ceiling of coordination maturation.

F6: Simple reaction time differs but complex reaction time shows no additional group difference. The shielded group is faster on simple RT and the group difference does not increase on choice RT, complex RT, or attention switching. The effect is a general speed advantage, not a timing-precision advantage. The timing channel hypothesis is not supported.

F7: Anticipatory timing precision (Task 4.3) is identical between groups after learning criterion is reached. Both groups learn the wheel-food pattern and produce anticipatory responses with equal temporal precision. Developmental RF exposure does not affect temporal encoding. This falsifies the temporal precision component of the hypothesis.

F8: Contextual fear conditioning temporal distribution (Task 4.5) is identical between groups. Both groups show identical freezing distributions around the conditioned interval. Temporal memory precision is unaffected by developmental RF exposure.

F9: Corticosterone levels differ between groups. Stress hormone differences indicate that the electromagnetic environments produced different stress levels. Any behavioral differences may be attributable to chronic stress rather than timing channel effects. The study has a confound.

Each criterion is specific, testable, and would genuinely terminate the relevant part of the hypothesis. A null result on F2 is the cleanest outcome and is published as a finding.

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7. Controls and Confounds

7.1 Acute versus chronic exposure

Testing occurs in a standard RF environment for both groups. Both groups experience identical acute RF during the 30-60 minute testing session. Any performance difference

is attributable to chronic developmental exposure history, not to acute presence or absence of RF during the test.

This is critical. If the shielded group were tested in a shielded room and the exposed group were tested in an exposed room, performance differences could reflect acute RF effects rather than developmental effects. Testing both groups in identical acute conditions isolates the developmental variable.

7.2 Stress verification

Corticosterone levels are measured from fecal samples at P28 and P56 in a subset of 20 mice per group. Fecal sampling is used rather than blood sampling to avoid handling stress confounds. If corticosterone levels differ between groups, the behavioral results are uninterpretable with respect to the timing channel hypothesis. This is falsification criterion F9.

7.3 Body weight and general health

Body weight is recorded weekly for all mice. Weekly health check by veterinary staff documents any signs of illness, injury, or abnormal behavior. If the shielded group is systematically heavier or healthier, behavioral differences may reflect general health rather than coordination precision.

7.4 Handling and transport

Same handlers for both groups, operating on the same schedule. Transport from home room to testing room follows identical protocols in identical opaque carriers along routes of identical length. Handling duration per session is logged and verified identical.

7.5 Estrous cycle

Female mice are tested without estrous cycle synchronization. Sex is included as a covariate in all analyses. If sex interacts with group, the interaction is reported and analyzed.

7.6 Litter effects

Pups from the same litter are distributed across testing subgroups where possible to avoid confounding litter effects with group effects. Litter is included as a random effect in all statistical models.

7.7 Equipment calibration

EMG electrodes verified at the start of each testing day against a known calibration signal. Rotorod RPM verified weekly with an optical tachometer. Force plate calibration verified daily with a known weight. Spectrum analyzer verification of RF levels in both rooms occurs daily for the duration of the study.

7.8 Faraday cage note

A commonly proposed control is to test balance inside a Faraday cage. This is invalid for the EMF coordination hypothesis as stated in HOWL-NEURO-1-2026. The coordination signal is endogenous — generated by the body's own tissue and propagating through the body's own tissue. A Faraday cage attenuates external fields entering or leaving the enclosure. It does not attenuate fields propagating within the body. The body is the medium. The cage is outside the medium. Testing inside a cage would block ambient RF but would not restore the developmental calibration that chronic exposure may have degraded. The developmental history is the variable, not the acute environment.

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8. Analysis Plan

8.1 Primary analysis

Mixed-effects linear model for each Tier 3 and Tier 4 task with the timing precision measure as dependent variable:

Dependent: bilateral onset latency SD (Tier 3), or task-specific timing measure (Tier 4)

Fixed effects: group (shielded vs exposed), developmental timepoint, group $\times$ timepoint interaction, sex

Random effect: litter

The group $\times$ timepoint interaction tests whether the developmental trajectory differs between groups. A significant main effect of group without interaction indicates a constant offset. A significant interaction indicates diverging trajectories.

8.2 Control analysis

The same model applied to Tier 1 tasks. The prediction is no significant effect of group. This analysis is performed first. If group effects appear on Tier 1 tasks, the Tier 3 and Tier 4 analyses are reported but interpreted with the caveat that a general effect is present.

8.3 Demand gradient analysis

A combined model including all tiers tests the tier $\times$ group interaction. The prediction is a significant positive tier $\times$ group interaction — the group difference increases with tier number. This is the demand gradient. If the interaction is absent, the effect is not demand-dependent and the timing channel hypothesis is not supported as the specific mechanism.

8.4 Developmental trajectory analysis

Nonlinear growth curve fitting (logistic or Gompertz) to timing precision across the six timepoints, separately by group. Parameters of interest: asymptote (ceiling precision),

inflection point (age at which half of mature precision is reached), and growth rate (slope at inflection). Group differences in each parameter are reported.

8.5 Reaction time decomposition

For the reaction time tasks (1.3, 2.3, 4.4), the processing stages are decomposed:

Simple RT = sensory detection + motor execution

Choice RT = sensory detection + discrimination + motor execution

Complex RT = sensory detection + discrimination + attention switch + motor execution

If the group difference is absent on simple RT and grows with each added processing stage, the timing channel is involved in inter-stage coordination, not in any single stage. This stage-additive pattern is the predicted signature.

8.6 Temporal precision analysis

For tasks with temporal encoding (4.3 and 4.5), the primary measure is the standard deviation of the temporally-relevant response across trials after learning criterion is reached. For Task 4.3: SD of anticipatory movement onset time relative to the transition point. For Task 4.5: SD of peak freezing time relative to expected shock time.

These measures are the purest tests of temporal encoding precision in the battery. They are analyzed separately from the motor coordination measures because they test a different prediction — that the timing channel is involved in temporal memory, not just motor coordination.

8.7 Multiple comparisons

The analysis plan specifies 14 primary dependent variables across Tiers 1-4. Bonferroni correction is applied: alpha per comparison = 0.05/14 = 0.0036. This is conservative. A false discovery rate approach (Benjamini-Hochberg) is reported as a secondary correction.

8.8 Null result reporting

If the primary analysis produces no significant group difference on any measure, the 95% confidence interval for the effect size is reported for each measure. These intervals bound the maximum effect size compatible with the null result and determine whether a larger study (more animals, more sensitive measures) could detect a smaller effect. The null result is published as a finding.

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9. Extended Predictions and Exploratory Analyses

The following analyses are labeled as exploratory. They are not part of the pre-registered primary analysis. They are included because they generate specific testable predictions

that would strengthen the interpretation of a positive primary result.

9.1 Learning curve shape

The timing channel hypothesis predicts that learning curves on novel motor tasks follow a specific pattern: initial rapid improvement (phase-lock network forming, timing channel establishing coordination patterns), plateau at an intermediate level (timing channel established but synaptic pathways still building), second improvement phase (synaptic pathways reaching maturity, content channel catching up to timing channel), final plateau (both channels mature).

This two-phase learning curve is predicted for both groups. The prediction is that the shielded group's first phase is steeper (faster phase-lock formation), the intermediate plateau is shorter (less time needed for the timing channel to stabilize), and the final plateau is higher (more precise timing foundation supports higher ceiling).

The exposed group is predicted to show a flatter first phase (slower phase-lock formation in noise), a longer intermediate plateau (more time needed for noisy timing channel to stabilize), and a lower final plateau (less precise timing foundation constrains ceiling).

9.2 Recovery from perturbation

On the rotorod, occasional perturbations can be introduced (brief speed changes, momentary stops and restarts). The recovery time from perturbation — time from perturbation onset to restoration of stable running — is predicted to be shorter in the shielded group. The perturbation recovery requires rapid re-establishment of inter-limb timing coordination after the disruption, which depends directly on the timing channel's precision.

9.3 Transfer between tasks

If a subset of mice is trained on one demanding task (e.g., accelerating rotorod) and then tested on a novel demanding task (e.g., narrow beam), the transfer between tasks tests whether the timing channel generalizes. The shielded group is predicted to show faster transfer — the general coordination precision transfers to novel contexts because the underlying timing network is more precisely calibrated.

9.4 Cross-frequency analysis of EMG

Spectral analysis of the EMG signal during demanding tasks may reveal frequency bands where bilateral coherence differs between groups. If the timing channel operates through phase-lock at specific frequencies, the bilateral EMG coherence at those frequencies will be higher in the shielded group. This exploratory analysis could identify the coordination frequencies and connect them to the theoretical predictions from HOWL-NEURO-1-2026.

9.5 Fatigue cliff characterization

HOWL-NEURO-1-2026 predicts that the two-channel model produces a fatigue cliff — stable performance followed by sudden collapse, rather than gradual degradation. On extended rotorod trials (longer duration at fixed speed), the shape of the performance degradation curve can be characterized. The timing channel hypothesis predicts a sharper cliff in both groups (this is a structural prediction, not a group-difference prediction), but an earlier cliff in the exposed group (the degraded timing channel fatigues sooner, loading the content channel earlier, which then also fatigues, producing compound failure earlier in the session).

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10. What This Study Does and Does Not Test

Does test: Whether chronic ambient RF exposure at residential WiFi power levels during the developmental period — from conception through adulthood — affects the precision of inter-muscle coordination timing, sensory-motor learning speed, and temporal encoding precision in mice.

Does not test: Whether WiFi causes disease, cancer, neurological damage, or gross motor impairment. The predicted effect is a subtle precision ceiling difference on demanding tasks, not impairment of basic function.

Does not test: Whether acute RF exposure affects motor performance. Both groups are tested in identical acute conditions.

Does not test: Whether the effect is specific to 2.4 GHz. A positive result motivates follow-up studies at other frequencies (5 GHz WiFi, cellular bands, Bluetooth) and at different power densities (dose-response). A negative result does not rule out effects at other frequencies.

Does not test: The mechanism. A positive result demonstrates that chronic developmental RF exposure affects motor timing precision. It does not determine whether the mechanism is direct EMF phase-lock interference, indirect effects on neural development, thermal microeffects, or something else. The mechanism question requires the frequency-specific interference experiments from HOWL-NEURO-1-2026.

Does not test: Whether the effect is reversible. A positive result motivates a follow-up cross-over study: raise mice in RF, move to shielded environment at P28, test at P84. If the timing precision partially recovers, the developmental effect is at least partially reversible. If it does not recover, the developmental window is critical and the effect is permanent.

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11. Cost and Feasibility

The study requires:

One RF-shielded room. Cost approximately $30,000-50,000 for construction if not available at the institution. Many institutions have RF-shielded rooms for EMC testing, MRI research, or electromagnetic compatibility work. Existing facilities may be adaptable.

Standard laboratory mouse housing. Available at any research university with an animal facility.

Standard motor testing equipment: rotorod, beam apparatus, open field arena, force plate, lever boxes, food dispensers. Available in any behavioral neuroscience facility.

Surface EMG recording equipment with 10 kHz sampling capability. Available in most motor physiology laboratories.

Spectrum analyzer for RF verification. Available in any EMF research laboratory or borrowed from an engineering department.

Video recording at standard frame rates for behavioral scoring. High-speed video (1,000+ fps) optional but valuable for landing and rapid reversal tasks.

Total duration: 12 weeks from mating to final testing at P84, plus 2 weeks for breeding setup and facility verification. Total: approximately 14 weeks of active study.

Total animals: 120, well within standard IACUC approval ranges for behavioral studies.

Estimated total cost excluding personnel: $50,000-80,000 if RF-shielded room must be constructed, $15,000-30,000 if existing shielded room is available.

The study is feasible at any research university with a behavioral neuroscience facility and access to RF-shielded space. The technical requirements are standard. The experimental design is straightforward. The analysis plan is pre-specified. The only non-standard element is the RF-shielded housing environment, and even this is adaptable from existing infrastructure at many institutions.

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12. Publication Commitment

Results are published regardless of outcome. Positive, negative, or ambiguous — the data is reported with the full analysis plan as pre-registered above.

The analysis plan is not modified after data collection. The primary analyses are performed exactly as specified. Any post-hoc analyses are labeled as exploratory and reported separately.

If the result is null: reported as a finding. Confidence intervals bound the maximum compatible effect size. The study contributes to the literature by establishing that chronic developmental RF exposure at residential WiFi levels does not produce a detectable effect on motor timing precision in mice at the study's statistical power.

If the result is positive: follow-up studies are proposed in the discussion. Frequency specificity (which RF bands), dose-response (what power density threshold), developmental

window (which period of exposure is critical), and reversibility (can the effect be remediated by moving to shielded environment at various ages).

If the result is ambiguous: trends that do not reach significance after correction are reported with their uncorrected p-values and effect sizes. The reader can judge whether a larger study is warranted.

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13. Connection to the Research Program

This study is one link in a chain that begins with the carrier elimination arithmetic of HOWL-NEURO-1-2026 and extends through the processing theory of HOWL-MATH-14-2026 to the soliton hierarchy of the PCTRM substrate model.

The carrier elimination establishes that EMF is necessary for sub-millisecond motor coordination. The processing theory provides the framework for understanding dissolution, cascade severity, and the developmental trajectory of coordination networks. The soliton hierarchy provides the model for understanding how the body's internal coordination channels interact with the electromagnetic environment of the parent soliton (Earth's technological electromagnetic output).

If this study produces a positive result, the next step is not another mouse study. The next step is the frequency-specific interference experiments from HOWL-NEURO-1-2026 — applying EMF at specific frequencies during specific tasks to identify the coordination bands. Once the coordination frequencies are identified, the question of whether they overlap with commercial wireless bands becomes a direct measurement rather than a theoretical prediction.

If this study produces a null result, the carrier elimination arithmetic still stands. The null would indicate that the endogenous coordination channel, if it exists, is robust to 2.4 GHz RF at residential power levels. This would be informative — it would constrain the architecture of the timing channel (ruling out broadband sensitivity, supporting frequency-specific or waveguide-protected architectures that are inherently shielded from out-of-band interference).

Either result advances the program. The study is designed to generate information regardless of direction.

References

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Table A: Testing Battery Summary

Task	Tier	Demand	Primary Measure	Control Purpose
Open field locomotion	1	Minimal	Distance, speed	General health/motivation
Wide beam (2 cm)	1	Low	Time, foot faults	Basic motor function
Simple reaction time	1	Minimal	RT in ms	Baseline speed control
Narrow beam (0.5 cm)	2	Moderate	Time, foot faults, falls	Balance at boundary demand
Fixed rotorod (15 RPM)	2	Moderate	Latency to fall	Sustained coordination at boundary
Choice reaction time	2	Moderate	RT, accuracy	Discrimination speed
Accelerating rotorod	3	High	RPM at fall	Coordination ceiling
Bipedal reach	3	Maximum	Hold time, success rate	Sustained coordination at maximum demand
Landing (30 cm drop)	3	High	Stabilization time, symmetry	Impact absorption timing
Rapid reversal	3	High	Reversal latency, overshoot	Motor program switching

Task	Tier	Demand	Primary Measure	Control Purpose
Sensory detection	4	Sensory	Detection threshold, RT	Sensory precision
Contingency learning	4	Cognitive-motor	Trials to criterion, reversal speed	Learning speed
Wheel-food-wheel-food	4	Temporal	Anticipatory latency, timing SD	Temporal pattern encoding
Complex reaction time	4	Multi-stage	Stage-specific RT, total RT	Inter-stage coordination
Fear conditioning timing	4	Temporal memory	Freezing distribution width	Temporal memory precision

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Table B: Predicted Outcomes by Tier

Tier	Prediction	If confirmed	If violated
1	No group difference	Controls hold, proceed to higher tiers	General confound, study compromised
2	Small or absent difference	Boundary region identified	Classical carriers may suffice at moderate demand
3	Measurable difference, shielded better	Coordination ceiling effect confirmed	Timing channel not demand-dependent
4	Measurable difference, shielded better	Sensory-motor integration and temporal encoding affected	Effect is motor-specific, not general timing

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Table C: Falsification Criteria Summary

ID	Criterion	What it means
F1	Tier 1 differs	General effect, not coordination-specific
F2	No Tier 3/4 difference at any age	No developmental RF effect on timing
F3	Uniform difference across tiers	General effect, not demand-dependent
F4	Timing differs, performance does not	Timing is not rate-limiting
F5	Identical developmental trajectories	No effect on maturation rate or ceiling
F6	Simple RT differs, complex RT shows no additional difference	General speed, not timing precision
F7	Anticipatory timing identical after learning	Temporal encoding unaffected
F8	Fear conditioning distribution identical	Temporal memory unaffected
F9	Corticosterone differs	Stress confound

Table D: Stage Decomposition of Reaction Time Tasks

Task	Stages	Predicted group difference
Simple RT (1.3)	Detection → Execution	None (control)
Choice RT (2.3)	Detection → Discrimination → Execution	Small, in discrimination stage
Complex RT (4.4)	Detection → Discrimination → Attention switch → Execution	Larger, cumulative across stages

If the group difference is additive across stages — each added stage contributes an additional timing advantage for the shielded group — the timing channel is involved in inter-stage coordination. If the group difference appears only in one stage, the timing channel is involved in that specific process.

Table E: Wheel-Food-Wheel-Food Protocol Detail

Phase	Duration	Event	Measure
Wheel access	120 seconds	Wheel unlocks	Running distance, speed
Transition 1	Instantaneous	Wheel locks, pellet dispenses	Latency from lock to food port approach
Food access	60 seconds	Food port open	Consumption time, food consumed
Transition 2	Instantaneous	Food port closes, wheel unlocks	Latency from close to wheel engagement
Repeat	8-12 cycles per session		

Learning criterion: Anticipatory food port approach (movement toward port beginning before pellet dispenses) on 3 consecutive cycles.

Timing precision measure (post-criterion): Standard deviation of anticipatory movement onset time relative to the transition point, across all post-criterion cycles.

Prediction: Shielded group reaches criterion in fewer cycles (faster temporal pattern learning) and shows lower onset time SD after criterion (more precise temporal encoding).

Table F: Cost Estimate

Item	Cost range	Notes
RF-shielded room construction	$30,000-50,000	If not available. Many institutions have existing shielded space.
WiFi access point	$50	Commercial router, standard model
Spectrum analyzer rental	$2,000-5,000	For daily RF verification. May be available in-house.
Animals (120 C57BL/6J)	$3,000-5,000	Including breeding pairs
Standard housing and husbandry	$5,000-8,000	14 weeks, 120 animals
Motor testing equipment	$0-15,000	Available in most behavioral neuroscience labs
EMG recording system	$0-10,000	Available in most motor physiology labs
Force plate	$0-5,000	Available in most biomechanics labs

Item	Cost range	Notes
Consumables (electrodes, chow, bedding)	$2,000-3,000	Standard supplies
Total (new shielded room)	**$50,000-80,000**	Excluding personnel
Total (existing shielded room)	**$15,000-30,000**	Excluding personnel

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END HOWL-NEURO-3-2026

Registry: HOWL-NEURO-3-2026

Status: Experimental proposal for independent implementation

Verification: All predictions pre-registered, all falsification criteria specified

Claims boundary: Proposal only. The author will not conduct this experiment. The design is published for any qualified laboratory to execute.

Falsification: Nine explicit criteria specified. Any single criterion met terminates the relevant part of the hypothesis.

The carrier elimination is arithmetic. The developmental prediction is a consequence. The experiment is a test. The results are published regardless of direction.

[**HOWL-NEURO-3-2026**] Developmental Motor Coordination in RF-Shielded Versus Standard Electromagnetic Environments. Github: https://github.com/ghowland/papers/tree/main/papers/NEURO/HOWL-NEURO-3-2026

[**HOWL-NEURO-1-2026**] Elimination of Classical Carriers for Sub-Millisecond Motor Coordination. Github: https://github.com/ghowland/papers/tree/main/papers/NEURO/HOWL-NEURO-1-2026

[**HOWL-NEURO-2-2026**] Fuel and Frame. Github: https://github.com/ghowland/papers/tree/main/papers/NEURO/HOW NEURO-2-2026